Lecture 14 - Oct 14

5.3 Unimodality of Binomial Theorem

Goal:

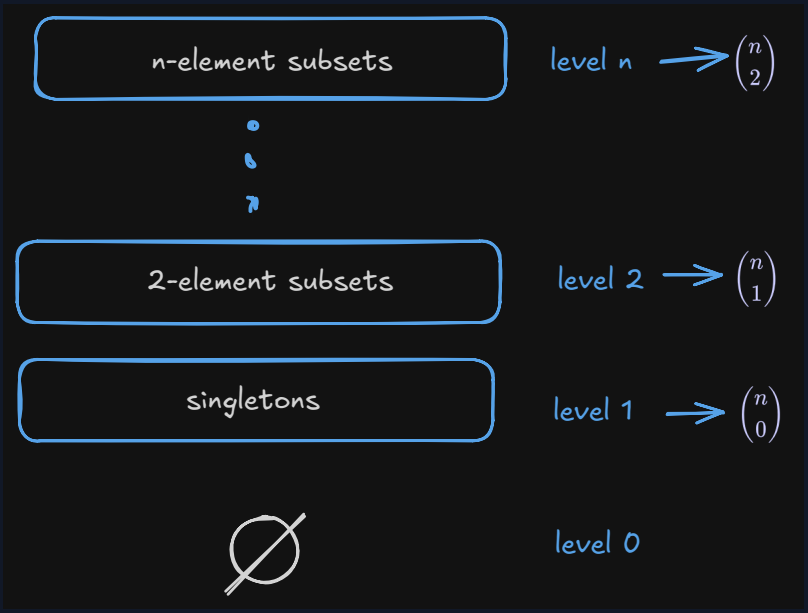
$$\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$$

which term is the largest?

Equivalent to asking if $P = (\mathcal{P}(S), \subseteq)$ is a poset.

So, what is the largest size of an antichain in P?

Consider the Hassel diagram of P:



Each level is an antichain

DEFINITION (UNIMODAL SEQUENCE).

The sequence $S_1, \ldots, S_n$ is **unimodal** provided that there exists an integer $t, 0 \leq t \leq n$, such that

$$S_0 \leq S_1 \leq \cdots \leq S_t \geq S_{t+1} \geq \ldots S_n$$

The possible choices for the value of t is $\binom{n+1}{1}$ since $0 \leq t \leq n$.

THEOREM (5.3.1 - BINOMAL COEFFICIENTS UNIMODALITY).

Let n be a positive integer. The sequence of binomal coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ is unimodal. More precisely,

$$\binom{n}{0} < \binom{n}{1} < \cdots < \binom{n}{\lfloor \frac{n}{2} \rfloor} \geq \binom{n}{\lceil \frac{n}{2} \rceil} > \cdots > \binom{n}{n}$$

$$\text{n is even, } \binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{n}{\lceil \frac{n}{2} \rceil}$$

$$\text{n is odd, } \binom{n}{\frac{n-1}{2}} = \binom{n}{\frac{n+1}{2}}$$

$$\binom{n}{k} = \binom{n}{n-k}$$

Recall the Lecture 12 - Oct 7 > Definition 9 (Maximal Chain).

For the poset $P = (\mathcal{P}(S), \subseteq)$, it can be seen that there are $n!$ possible maximal chains:

$$\emptyset, \{1\}, \{1,2\}, \dots, \{1,2,\dots n\}$$

At each step an element must get picked, and the order in which they get picked matters, making there be $n!$ possible such chains. All maximal chains must be of this form, meaning of size $n+1$.

THEOREM (SPERNER 1928).

Let $S = [n]$. Then, every antichain in $P = (\mathcal{P}(S), \subseteq)$ contains at most $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subsets of S.

Question: If A is a subset of $[n]$ and $P = (\mathcal{P}([n]), \subseteq)$, How many maximal chains in P containing A?
is it $\leq |A|$?

Proof.@Theorem 2 (5.3.1 - Binomal Coefficients Unimodality)

$$\frac{\binom{n+1}{k}}{\binom{n}{k}} = \frac{\frac{n!}{(k+1)!(n-k-1)!}}{\frac{n!}{k!(n-k-1)!}} = \frac{n-1}{k+1}$$

So, we must find when this ratio $< 1, = 1, > 1$, Which would give an indication of which are bigger.

TODO: finish proof?

□

5.6 - Symmetric chain partition

for Sperner's theorem

Let $S = [n], P = (\mathcal{P}(S), \subseteq)$.

DEFINITION (SYMMETRIC CHAIN PARTITION).

A chain partition of P is a **symmetric chain partition** if for each chain $C_1, C_2, \ldots, C_m$ in the partition:

1. $C_1 \subseteq C_2 \subseteq \cdots \subseteq C_m$
2. $|C_i| = |C_{i+1}| - 1 \iff |C_i| + 1 = |C_{i+1}|$ for all $1 \leq i \leq m-1$
3. $|C_1| + |C_m| = n$

Example

Find a symmetric chain partition of $P = (\mathcal{P}(S), \subseteq)$, where $S = \{1, 2, 3\}$

Ans:

$n = 3$, odd, so $\binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{3}{1} = 3$. So, we must have 3 chains in the partition.

$$\begin{aligned} \emptyset, \{1\}, \{1,2\}, \{1,2,3\} &\rightarrow \text{maximal chain} \\ \{2\}, \{2,3\} & \\ \{3\}, \{1,3\} & \end{aligned}$$

Remark: For the symmetric chain partition, the $\emptyset$ must be part of the maximal chain in order to make the partition symmetric.

Finding symmetric chain partitions

Question: How many chains are there in a symmetric chain partition?

Ans:

Sperner's Theorem: at most $\binom{n}{\lfloor \frac{n}{2} \rfloor}$.

Dilworth's Theorem: at least $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ because the level $|L_{\lfloor \frac{n}{2} \rfloor}| = \binom{n}{\lfloor \frac{n}{2} \rfloor}$.

So,

$$\binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{n}{\lceil \frac{n}{2} \rceil}$$

Question: How to obtain a symmetric chain partition for $P = (\mathcal{P}([n]), \subseteq)$?

Answer: Obtained inductively from a symmetric chain partition for $(\mathcal{P}([n-1]), \subseteq)$.

For each chain $C_1, \ldots, C_m$ from the poset with $S = [n-1]$, it is known that

$$\begin{aligned} C_1 \subset C_2 \subset \cdots \subset C_m, & \\ |C_1| + |C_m| = n - 1 & \\ |C_i| + 1 = |C_{i+1}|, 1 \leq i \leq n-2 & \end{aligned}$$

We create one/two new chains for the $[n]$ partition from the chain of the $[n-1]$ partition.

If $m \geq 2$, then two get created:

$$C_1 \subset C_2 \subset \cdots \subset C_{m-1} \subset C_m \cup \{n\}$$
$$\to |C_1| + |C_m \cup \{n\}| = n = |S|$$

$$C_1 \cup \{n\}, C_2 \cup \{n\}, \ldots, C_{m-1} \cup \{n\}$$
$$\to |C_1 \cup \{n\}| + |C_{m-1} \cup \{n\}| = |C_1| + |C_{m-1}| + 2 = n$$

The $m = 1$, is a bit different

$$m = 1 \to |C_m| + |C_m| = n - 1$$
$$2|C_m| = n - 1 \implies n \text{ is odd}$$
$$= |C_m| = \frac{n-1}{2}$$

So, to create a new chain:

$$C_m \subset C_m \cup \{n\}$$
$$|C_m| + |C_m \cup \{n\}| = 2|C_m| + 1 = (n-1) + 1 = n$$

Example

Find a symmetric chain partition of $P = (\mathcal{P}(S), \subseteq)$, where $S = \{1, 2, 3, 4\}$

We know for $S = \{1, 2, 3\}$:

$$\emptyset \subset \{1\} \subset \{1, 2\} \subset \{1, 2, 3\}$$
$$\{2\} \subset \{2, 3\}$$
$$\{3\} \subset \{1, 3\}$$

For $S = [4]$, we can find the expected # of chains with Sperner's:

$$n = 4, \binom{4}{\lfloor \frac{4}{2} \rfloor} = \binom{4}{2} = 6$$

The chains become:

$$\emptyset \subset \{1\} \subset \{1, 2\} \subset \{1, 2, 3\} \subset \{1, 2, 3, 4\}$$
$$\{4\} \subset \{1, 4\} \subset \{1, 2, 4\}$$

$$\{2\} \subset \{2, 3\} \subset \{2, 3, 4\}$$
$$\{2, 4\}$$

$$\{3\} \subset \{1, 3\} \subset \{1, 3, 4\}$$
$$\{3, 4\}$$

For $S = [5]$,

$$n = 5, \binom{5}{\lfloor \frac{5}{2} \rfloor} = \binom{5}{2} = \frac{5!}{2!3!} = 10$$

So the chains are:

$$\emptyset \subset \{1\} \subset \{1,2\} \subset \{1,2,3\} \subset \{1,2,3,4\} \subset \{1,2,3,4,5\}$$
$$\{5\} \subset \{1,5\} \subset \{1,2,5\} \subset \{1,2,3,5\}$$

$$\{4\} \subset \{1,4\} \subset \{1,2,4\} \subset \{1,2,4,5\}$$
$$\{4,5\} \subset \{1,4,5\}$$

$$\{2\} \subset \{2,3\} \subset \{2,3,4\} \subset \{2,3,4,5\}$$
$$\{2,5\} \subset \{2,3,5\}$$

$$\{2,4\} \subset \{2,4,5\}$$

$$\{3\} \subset \{1,3\} \subset \{1,3,4\} \subset \{1,3,4,5\}$$
$$\{3,5\} \subset \{1,3,5\}$$

$$\{3,4\} \subset \{3,4,5\}$$